

HIGGS Boson Classification Using Advanced Machine Learning Pipelines

Machine Learning Final Project Report

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Abstract

This project applies an end-to-end machine learning workflow to the massive HIGGS dataset to solve a critical scientific and business analytics challenge: automating the identification of Higgs Boson events from particle collision data. Using a dataset of over 10.9 million observations generated via Monte Carlo simulations, a rigorous pipeline involving data cleaning, duplicate removal, feature scaling, and model exploration was implemented. Four distinct classification algorithms—Logistic Regression, Decision Trees, Random Forest, and XGBoost—were developed and evaluated. XGBoost achieved the highest predictive performance, delivering an Accuracy of **73.70%** and an F1-score of **76.22%**. Hyperparameter tuning via RandomizedSearchCV further optimized the model, ensuring generalization and reducing operational risks. The results demonstrate that advanced ensemble learning methods can effectively bypass manual feature engineering, driving faster and more reliable scientific discoveries.

1. Problem Definition

1.1 Objective and Scope

In modern particle physics, identifying rare subatomic particles like the Higgs Boson requires processing astronomical amounts of data generated by particle accelerators. The core objective of this project is to develop a highly accurate binary classification model that can distinguish between two distinct states:

- **Signal:** Particle collision events that produce a Higgs Boson.
- **Background:** Standard physics processes that mimic the signal but do not contain the Higgs Boson.

1.2 Business and Scientific Value

Traditionally, physicists have had to manually derive complex, high-level mathematical features from raw detector data to isolate these signals. This process is time-consuming, requires intense domain expertise, and is prone to human bias. By formulating this as a machine learning task, we aim to automate the detection process.

For decision-makers and research institutions, a highly accurate predictive model reduces manual analysis effort, optimizes computational resources, and significantly accelerates the pace of scientific discovery.

2. Data Collection & Description

2.1 Data Source and Licensing

The dataset used for this analysis is the **HIGGS Dataset**, sourced from the reputable **UCI Machine Learning Repository**. The data is publicly available for academic and research purposes, ensuring full compliance with licensing guidelines.

2.2 Dataset Characteristics

The dataset is exceptionally large, making it highly suitable for deep learning and advanced machine learning frameworks.

- **Total Observations:** 10,999,999
- **Total Variables:** 29 (1 Target Column + 28 Predictive Features)
- **Data Origin:** Produced using high-fidelity Monte Carlo simulations.
- **Feature Split: Low-Level Features (Columns 2–22):** 21 kinematic properties measured directly by the particle detectors in the accelerator (e.g., momenta, angles).
- **High-Level Features (Columns 23–29):** 7 functions derived by physicists from the low-level features to help separate signal from background.

3. Data Preprocessing

A thorough preprocessing pipeline was executed to guarantee data integrity and optimize the training performance of the models.

- **Column Renaming:** To improve readability and workflow management within the Python notebook, all feature columns were assigned standardized names differentiating low-level detector data from high-level physics metrics.
- **Missing Value Analysis:** A programmatic scan for null values returned **zero missing values** across all 10,999,999 rows. This is expected from high-quality Monte Carlo simulation outputs.
- **Deduplication (Handling Duplicates):** Duplicate analysis identified **278,698 duplicate records**. Leaving these in would cause data leakage and artificially inflate model metrics. These rows were successfully removed, leaving a robust dataset of **10,721,301 unique observations**.
- **Feature Scaling:** Linear and distance-based algorithms like Logistic Regression are highly sensitive to variable scales. To address this, **StandardScaler** was applied to normalize the feature distributions, shifting the mean to \$0\$ and scaling variance to 1.
- **Data Splitting:** In alignment with the original dataset documentation, the data was stratified and split to ensure a rigorous evaluation, reserving a designated test partition (including the final 500,000 rows benchmark) to evaluate real-world generalization.

4. Exploratory Data Analysis

The descriptive and exploratory data analysis provided vital structural understandings of the feature ecosystem:

- **Class Distribution:** The target variable shows a relatively balanced distribution between the signal (1) and background (0) events. This balance is ideal for binary classification as it eliminates the need for advanced resampling techniques (like SMOTE) and ensures metrics like Accuracy remain highly reliable.
- **Statistical Summary:** Evaluation of the feature ranges via `df.describe()` showed that the variables are already structured and mostly bounded, with minimal presence of extreme, malicious outliers.
- **Correlation Structure:** The EDA confirmed that while low-level kinematic features share complex, non-linear relationships, the high-level features show strong correlations with specific low-level combinations. This validates the physicists' manual efforts but also opens the door for machine learning to map these relationships automatically.

5. Model Building

To solve this classification problem, four distinct machine learning algorithms representing different structural frameworks were selected:

1. **Logistic Regression:** Selected as the baseline linear classifier. It assumes a linear relationship between log-odds of the target and the features. It is fast, highly interpretable, and establishes a foundational performance standard.
2. **Decision Tree:** A non-linear, non-parametric model that builds a flowchart-like tree structure. It was selected because it naturally captures feature interactions without requiring explicit non-linear transformations.
3. **Random Forest:** An ensemble bagging method that builds multiple decision trees on random subsets of data and averages their predictions. This was selected to reduce the high variance and overfitting tendencies inherent to single decision trees.
4. **XGBoost:** A state-of-the-art sequential gradient boosting algorithm. It corrects the errors of previous trees iteratively. It was selected because it handles massive datasets exceptionally well, features built-in regularization, and consistently delivers top-tier predictive power on tabular data.

6. Model Evaluation & Comparison

All four models were tested on the unseen test partition. Performance was evaluated using four major metrics: Accuracy, Precision, Recall, and F1-Score.

6.1 Performance Matrix Table

The empirical results compiled from the Python notebook outputs are summarized below:

Algorithm Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	64.58%	64.53%	77.61%	70.47%
Decision Tree	63.97%	66.99%	66.69%	66.84%
Random Forest	73.08%	74.51%	76.85%	75.66%
XGBoost (Best Model)	73.70%	75.07%	77.41%	76.22%

6.2 Comparative Analysis

- **Linear vs. Non-Linear Performance:** Logistic Regression performed poorly in accuracy (64.58%) compared to ensemble methods. This proves that particle collision data possesses complex, non-linear relationships that linear models cannot fully map.
- **The Power of Ensembles:** Both Random Forest (73.08%) and XGBoost (73.70%) showed a significant leap in performance. By aggregating multiple estimators, they successfully filtered out background noise.
- **The Winning Model: XGBoost** emerged as the superior model. It achieved the highest overall balance with a **73.70% Accuracy** and a **76.22% F1-Score**, showing its superiority in handling complex simulated physics distributions.

7. Hyperparameter Tuning

Given its superior performance, **XGBoost** was selected for hyperparameter optimization. Due to the massive volume of observations (10.7), GridSearchCV was computationally unfeasible. Hence, RandomizedSearchCV was utilized to scan the parameter space efficiently.

7.1 Tuned Parameters & Justification

- **n_estimators (Tuned to 100):** Controls the number of sequential trees. Selected to provide enough boosting rounds without inducing excessive training time.
- **max_depth (Tuned to 6):** Limits the maximum depth of a tree. Selected to prevent deep trees from memorizing noise (overfitting).
- **learning_rate (Tuned to 0.1):** The step size shrinkage used to prevent overfitting during weights update. A value of 0.1 ensures stable, robust convergence.
- **subsample (Tuned to 0.8):** Specifies that 80% of the data is randomly sampled before building each tree, adding a layer of randomization to fight variance.

7.2 Results and Impact

Tuning successfully optimized the internal bias-variance tradeoff of the XGBoost framework. It provided tighter confidence intervals, minimized the gap between training and testing performance, and ensured that the model generalizes robustly to real-world physics streams rather than overfitting to simulation quirks.

8. Interpretation & Business Insights

Translating numerical metrics into actionable operational value is the core value of this pipeline.

8.1 Scientific and Operational Implications

- **The Power of High Recall (77.41%):** In particle physics, missing a true Higgs Boson event (a False Negative) is a massive loss of scientific opportunity. XGBoost's high recall ensures that the model successfully catches the vast majority of true signal events, ensuring rare data is not discarded.
- **The Impact of High Precision (75.07%):** High precision ensures that when the model flags a collision as a "Signal", it is highly likely to be correct. This drastically minimizes "False Alarms", saving research institutions thousands of hours of manual verification and downstream computational verification costs.

8.2 Strategic Alignment (Obviating Manual Engineering)

The performance achieved by XGBoost proves the project's core hypothesis: advanced machine learning frameworks can successfully learn complex subatomic relationships directly from low-level raw detector metrics. This effectively minimizes the absolute necessity for human physicists to spend years manually creating derived high-level features.

8.3 Limitations, Ethical, and Operational Risks

- **Simulation vs. Real World Bias:** The model is trained on Monte Carlo simulations. If real-world particle accelerators introduce unexpected hardware noise or structural anomalies not captured by simulations, the model's accuracy could degrade.
- **Computational Overhead:** Ensembles like XGBoost require significant memory and GPU infrastructure to train and run at scale. Deploying this model into a live, real-time data stream at an accelerator requires balancing inference latency with predictive accuracy.
- **Lack of Direct Explainability:** Advanced tree boosting operates as a semi-black box. In critical scientific research, not knowing the exact mathematical equation behind a classification can slow down peer-reviewed validation. This risk can be mitigated in future phases using XAI (Explainable AI) frameworks like SHAP values.